**“Protocol & Definitions” note**:

- horizon / overlap mapping

1)Horizon(how far ahead the risk forecast is made) for both cases step size 1 and 10

2)what we are doing :

**Case A — Step size = 1(refit every day)** Window length = 1000,

**Forecast horizon** = **1 day**

| | | |
|---|---|---|
| Trained on day 1 to day 1000 | Predicted var on day 1001 | They are non overlapping so all tests are valid |
| Refitted on day 2 to day 1001 | Predicted var on day 1002 | |
| Refitted on day 3 to day1002 | Predicted var on day 1003 | |

**Forecast horizon = 10 days** Here we simulate 10000 paths on day 1 and using this we will get day 1 returns . with these returns we will update mu and sigma for next day and again we simulate 10000 paths , then continue the same process for 10 days which will be added so that we will get 10000 (10 days returns) and compared with cumulative sum of actual returns(log) on 10 days for VaR and ES

| | | |
|---|---|---|
| Trained on day 1 to day 1000 | Predicted var on day 1001 to day 1010 | They are overlapping so christophersson and dq,kupeic,ascerby es test tests are not used |
| Refitted on day 2 to day 1001 | Predicted var on day 1002 to day 1011 | |
| Refitted on day 3 to day1002 | Predicted var on day 1003 to day 1012 | |

**Case b — Step size = 10** Window length = 1000,

Forecast horizon = **1 day**

| | | |
|---|---|---|
| Trained on day 1 to day 1000 | Predicted var on day 1001 | They are non overlapping so all tests are valid |

| | | |
|---|---|---|
| Refitted on day 11 to day 1010 | Predicted var on day 1011 | |
| Refitted on day 21 to day1020 | Predicted var on day 1021 | |

Forecast horizon = **10 day**

| | | |
|---|---|---|
| Trained on day 1 to day 1000 | Predicted var on day 1001 to day 1010 | They are non overlapping so all tests are valid |
| Refitted on day 11 to day 1010 | Predicted var on day 1011 to day 1020 | |
| Refitted on day 21 to day1020 | Predicted var on day 1021 to day 1030 | |

for these tests we are using violations data(where we compare forecast with realized return) and as long as the forecasted days are not overlapping even though training data is overlapping we can consider these tests(backtests) as valid . By the word overlap we should say it based on forecasted days not on training days .

- P&L construction,

Here for horizon 1 we used daily log returns but for horizon 10 even though we are using daily log returns for fitting but on predicted days we will predict for 10 days and add these 10 values and compare with actual 10 days sum of returns

- exceedance-day shortfall error,

I am simulating 10000 paths so we have 10000 probable returns finally, now i have calculated var by considering these values and i have calculated ES by taking mean of values lesser than var. exceedance-day shortfall error is calculated by taking mean of absolute difference of ES predicted with actual return on the var violation days .

- which backtests are considered valid under each scenario.

For step size 1 and horizon 10 case christopherson and dq tests are not applicable and also for kupeic as the forecast window is overlapping, so violations will also overlap which will result in more frequency of violations, ascerby es test also includes assumption of independence which means it is not better to consider these tests as well . but for remaining cases all are applicable .

For backtesting i will be using Kupeic, Christophersson independence , DQ test with lags 1 and 4, Acerby ES test , and one scoring rule that is FZ0 Loss(applicable for all cases)

**Results:** https://drive.google.com/drive/folders/1CyfqGuNjzUM1OexK9EfI2tqNtI1o8yHv?usp=sharing

**Code link :**
https://drive.google.com/drive/folders/1gyRWqAHvmaT4BdI0BKn5vPQheEhejWxd?usp=sharing

**For step size 10 we have 248 observations(window is refitted 248 times )**

**For step size 1 we have 2473 observations**

**Sparse means threshold passes all tests but exceedences are less than 100**

**Invalid means threshold not passes some tests**

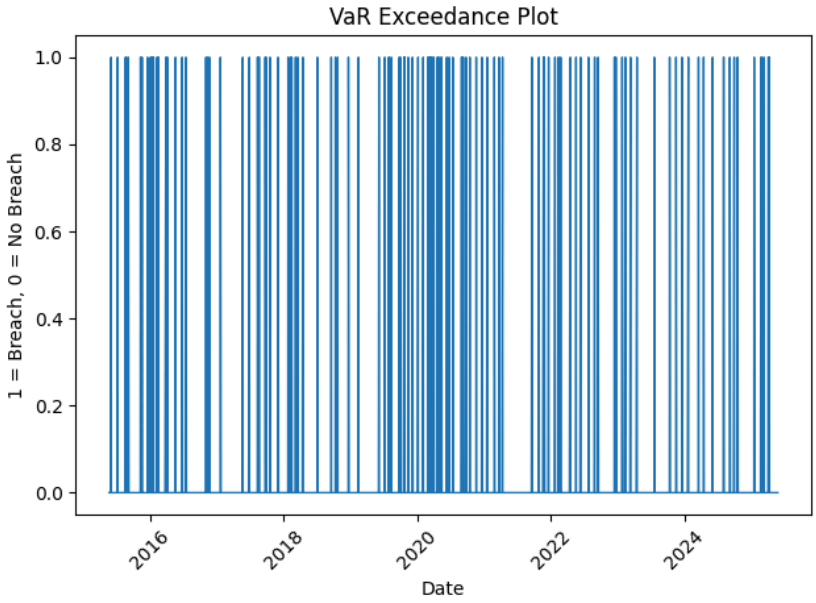
**Lower total tells how many times threshold fails in lower tails**

1) **Horizon 1 No KDE on Tails**

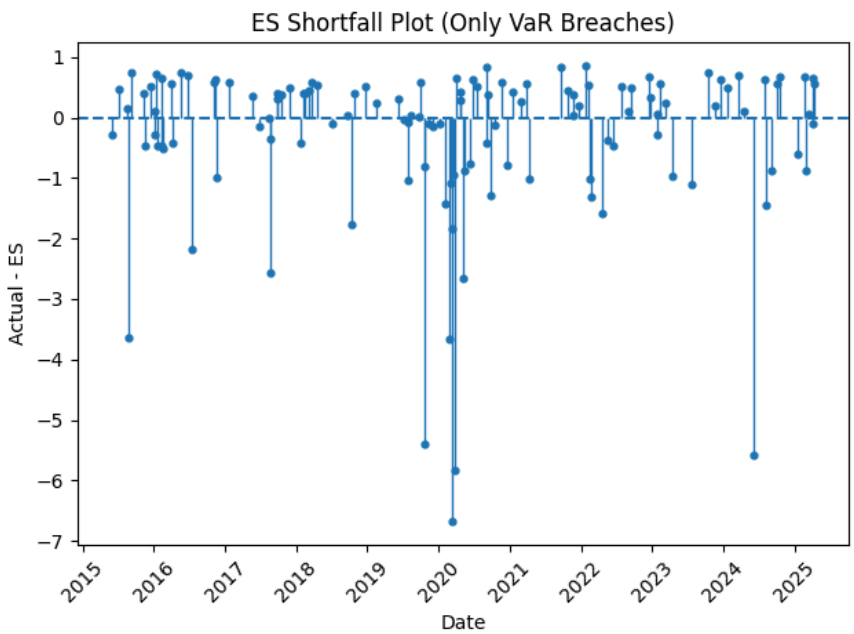
**a)Step Size 1 and 2 assets**

| Count of Reason | Column Labels | | | | | | |
|---|---|---|---|---|---|---|---|
| | lower | | | lower Total | upper | | | upper Total | Grand Total |
| Row Labels | invalid | sparse | (blank) | | invalid | sparse | (blank) | | |
| INFY.NS | | | | | 12 | 93 | | 105 | 105 |
| SBIN.NS | 435 | 152 | | 587 | | 37 | | 37 | 624 |
| **Grand Total** | **435** | **152** | | **587** | **12** | **130** | | **142** | **729** |

**Exceedence Plot**



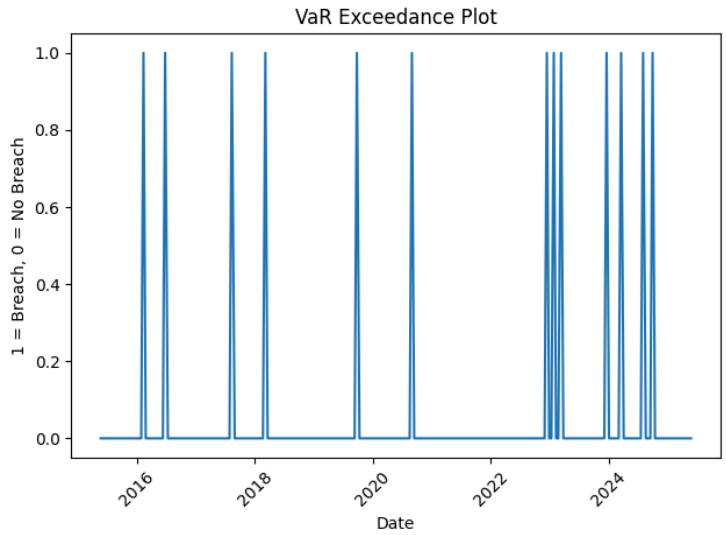
**ES Shortfall plot**



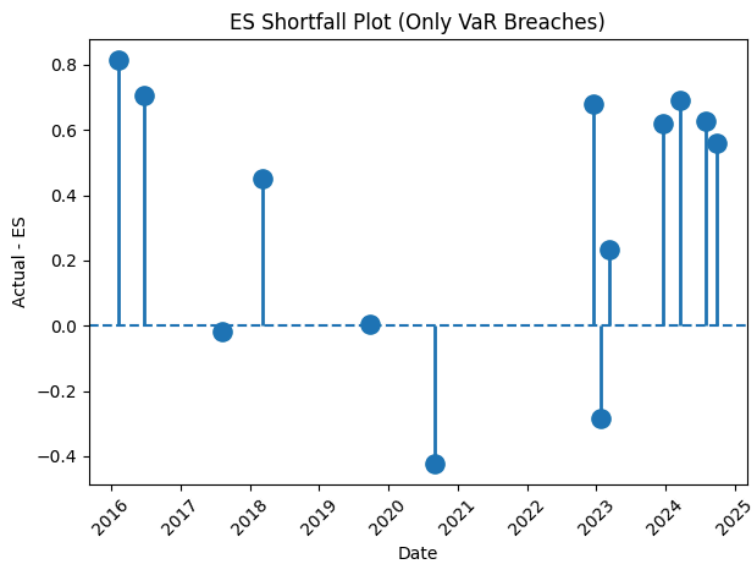

**b)Step size 10 and 2 assets**

| Count of Reason | Column Labels | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | lower | | | lower Total | upper | | | upper Total | Grand Total |
| Row Labels | invalid | sparse | (blank) | | invalid | sparse | (blank) | | |
| INFY.NS | | | | | 1 | 10 | | 11 | 11 |
| SBIN.NS | 43 | 15 | | 58 | | 4 | | 4 | 62 |
| **Grand Total** | **43** | **15** | | **58** | **1** | **14** | | **15** | **73** |

**Exceedence Plot**



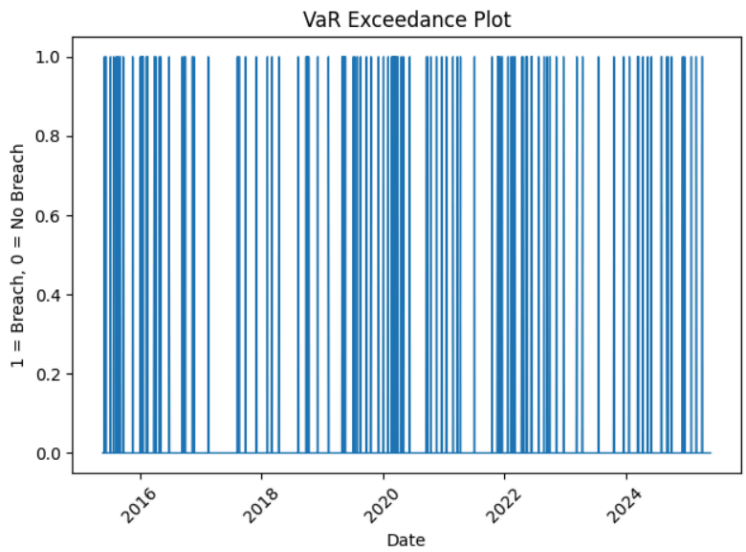

## ES Shortfall plot



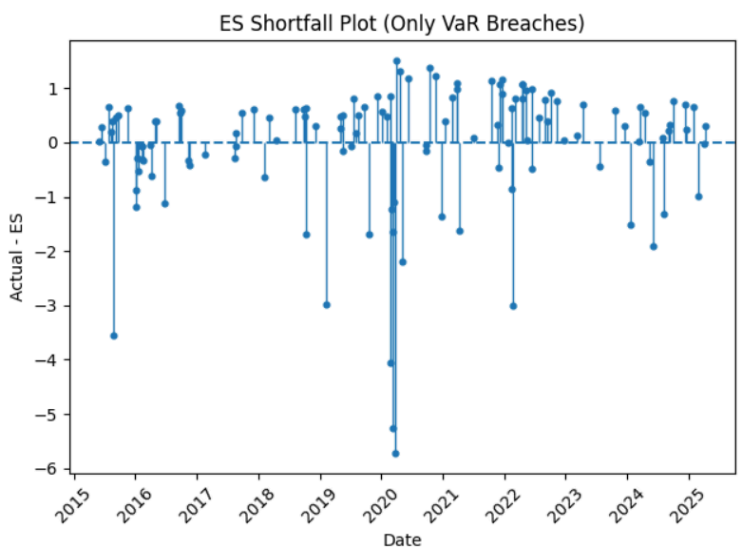

## c) Step size 1 and 4 assets

| Count of Reason | Column Labels | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | lower | | | lower Total | upper | | | upper Total | Grand Total |
| Row Labels | invalid | sparse | (blank) | | invalid | sparse | (blank) | | |
| HINDUNILVR.NS | 535 | 1288 | | 1823 | | 344 | | 344 | 2167 |
| INFY.NS | | | | | 13 | 95 | | 108 | 108 |
| SBIN.NS | 435 | 152 | | 587 | | 37 | | 37 | 624 |
| TATAMOTORS.NS | 68 | 111 | | 179 | 24 | 741 | | 765 | 944 |
| **Grand Total** | **1038** | **1551** | | **2589** | **37** | **1217** | | **1254** | **3843** |

## Exceedence Plot



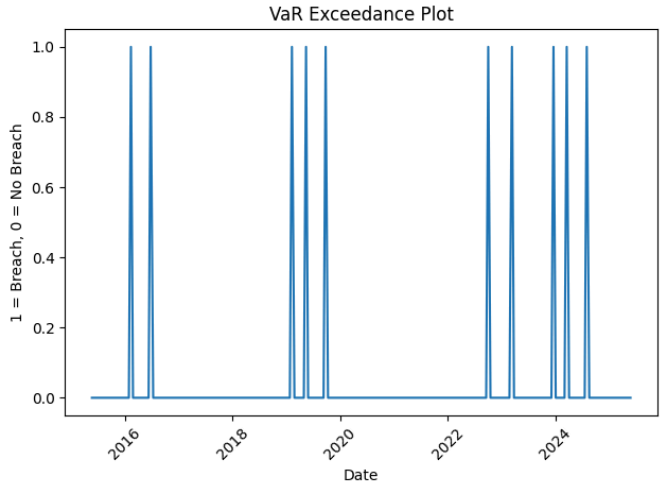

## ES Shortfall plot



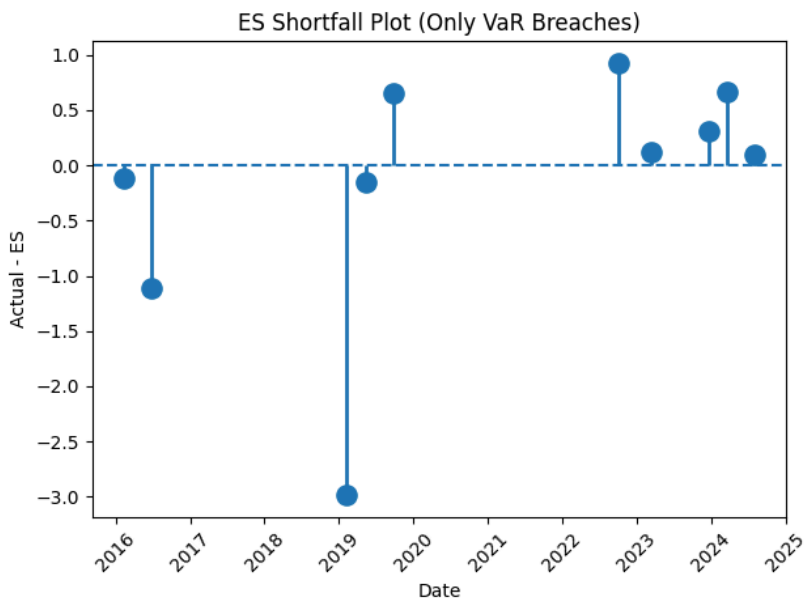

## d) Step size 10 and 4 assets

| Count of Reason | Column Labels | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | lower | | | lower Total | upper | | | upper Total | Grand Total |
| Row Labels | invalid | sparse | (blank) | | invalid | sparse | (blank) | | |
| HINDUNILVR.NS | 52 | 131 | | 183 | | 34 | | 34 | 217 |
| NFY.NS | | | | | 1 | 10 | | 11 | 11 |
| SBIN.NS | 43 | 15 | | 58 | | 4 | | 4 | 62 |
| TATAMOTORS.NS | 10 | 13 | | 23 | 4 | 74 | | 78 | 101 |
| **Grand Total** | **105** | **159** | | **264** | **5** | **122** | | **127** | **391** |

**Exceedence Plot**




**ES Shortfall plot**

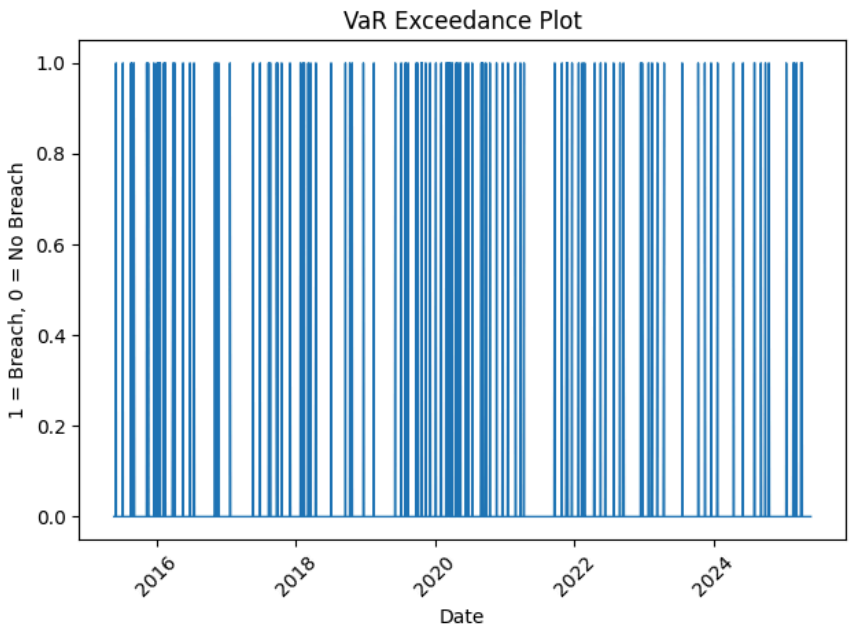



**2) Horizon 1 with KDE on Tails**

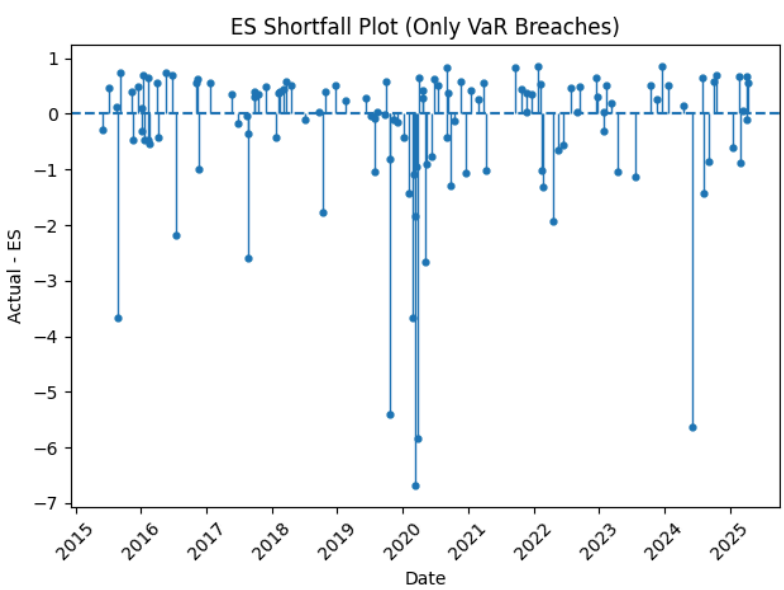
**a)Step Size 1 and 2 assets**

| Count of Reason | Column Labels | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | lower | | | lower Total | upper | | | upper Total | Grand Total |
| Row Labels | invalid | sparse | (blank) | | invalid | sparse | (blank) | | |
| INFY.NS | | | | | 12 | 93 | | 105 | 105 |
| SBIN.NS | 435 | 152 | | 587 | | 37 | | 37 | 624 |
| **Grand Total** | **435** | **152** | | **587** | **12** | **130** | | **142** | **729** |

## Exceedence Plot



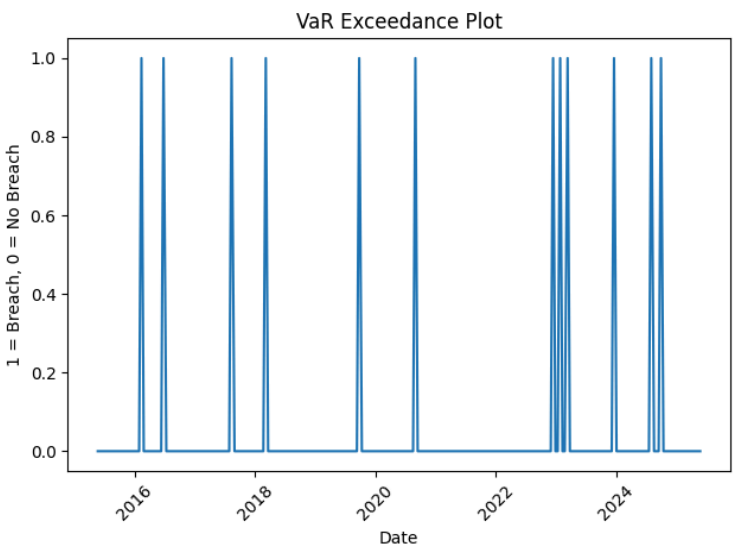

## ES Shortfall plot



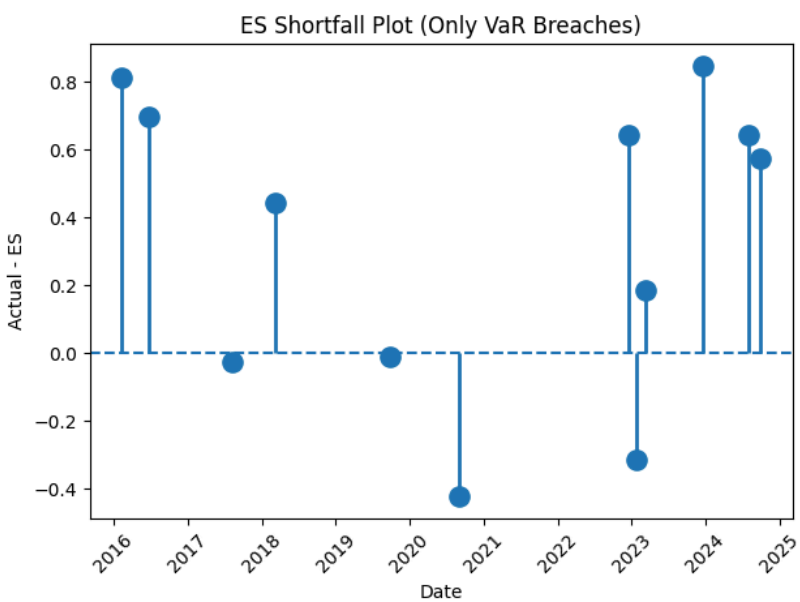
**b)Step size 10 and 2 assets**

| Count of Reason | Column Labels | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | lower | | | lower Total | upper | | | upper Total | Grand Total |
| Row Labels | invalid | sparse | (blank) | | invalid | sparse | (blank) | | |
| INFY.NS | | | | | 1 | 10 | | 11 | 11 |
| SBIN.NS | 43 | 15 | | 58 | | 4 | | 4 | 62 |
| **Grand Total** | **43** | **15** | | **58** | **1** | **14** | | **15** | **73** |

**Exceedence Plot**



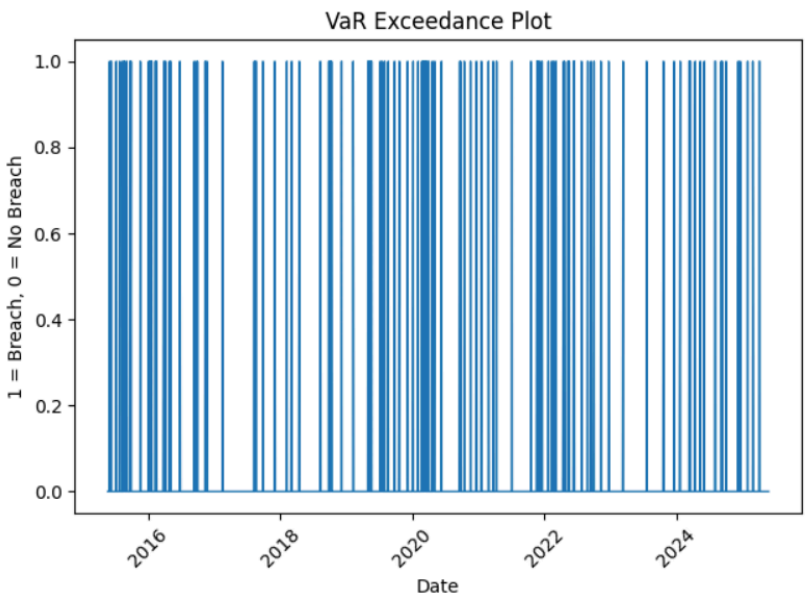

**ES Shortfall plot**



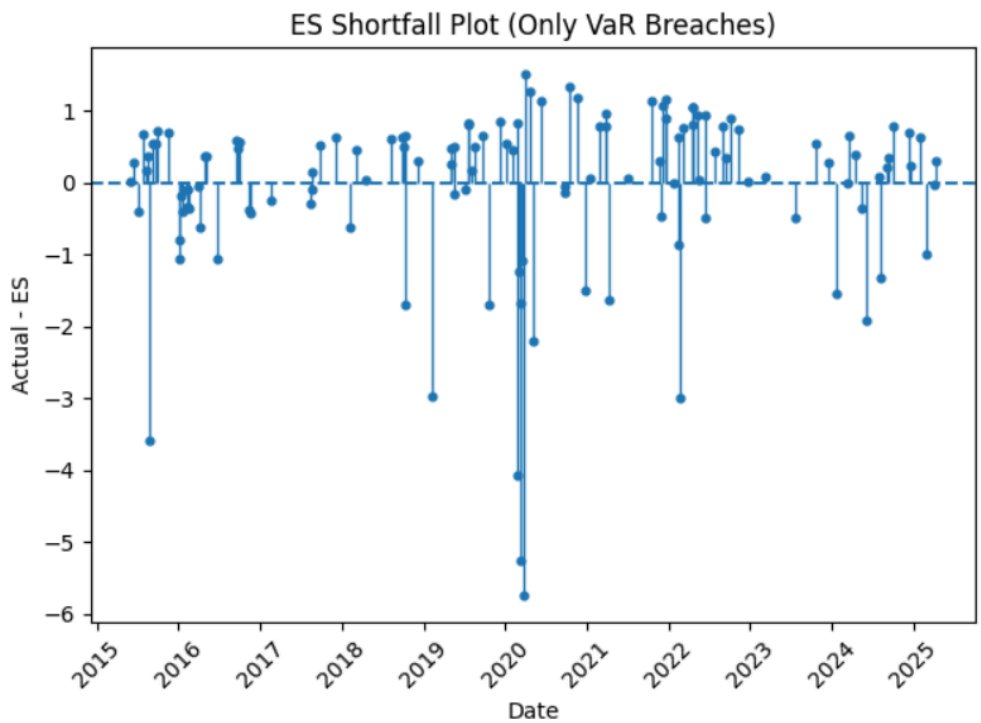
### c) Step size 1 and 4 assets

| Count of Reason | Column Labels | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | lower | | | lower Total | upper | | | upper Total | Grand Total |
| Row Labels | invalid | sparse | (blank) | | invalid | sparse | (blank) | | |
| HINDUNILVR.NS | 535 | 1288 | | 1823 | | 344 | | 344 | 2167 |
| INFY.NS | | | | | 13 | 95 | | 108 | 108 |
| SBIN.NS | 435 | 152 | | 587 | | 37 | | 37 | 624 |
| TATAMOTORS.NS | 68 | 111 | | 179 | 24 | 741 | | 765 | 944 |
| **Grand Total** | **1038** | **1551** | | **2589** | **37** | **1217** | | **1254** | **3843** |

### Exceedence Plot



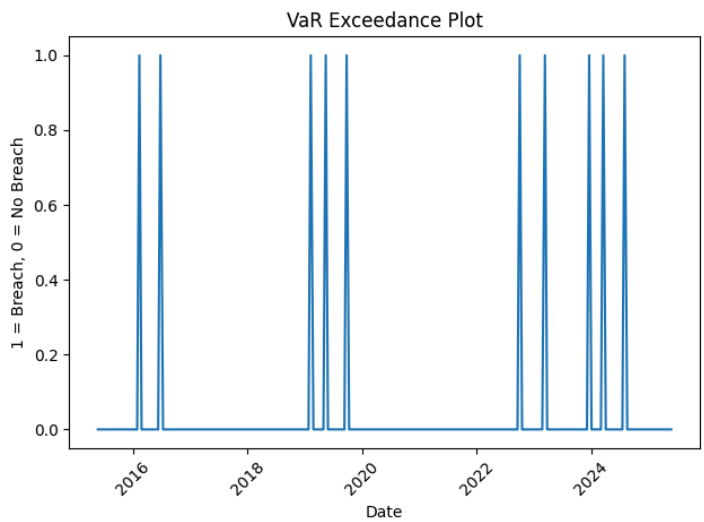

### ES Shortfall plot



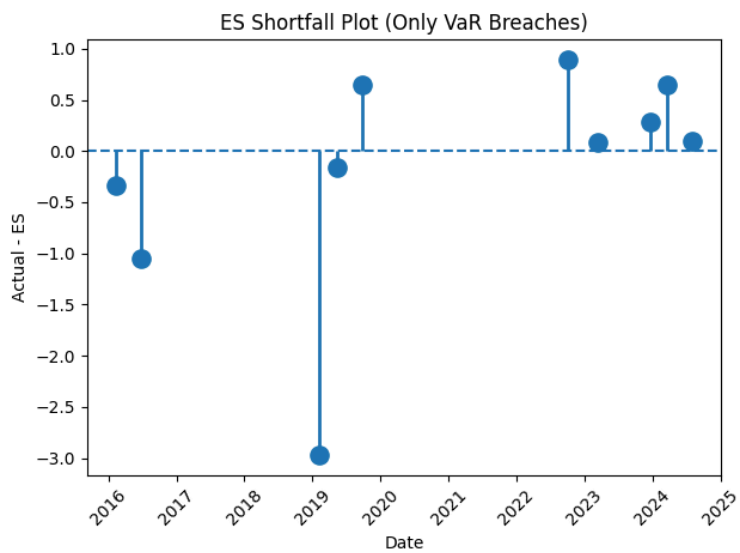

**d) Step size 10 and 4 assets**

| Count of Reason | Column Labels | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | lower | | | lower Total | upper | | | upper Total | Grand Total |
| Row Labels | invalid | sparse | (blank) | | invalid | sparse | (blank) | | |
| HINDUNILVR.NS | 52 | 131 | | 183 | | 34 | | 34 | 217 |
| INFY.NS | | | | | 1 | 10 | | 11 | 11 |
| SBIN.NS | 43 | 15 | | 58 | | 4 | | 4 | 62 |
| TATAMOTORS.NS | 10 | 13 | | 23 | 4 | 74 | | 78 | 101 |
| **Grand Total** | **105** | **159** | | **264** | **5** | **122** | | **127** | **391** |

**Exceedence Plot**




**ES Shortfall plot**

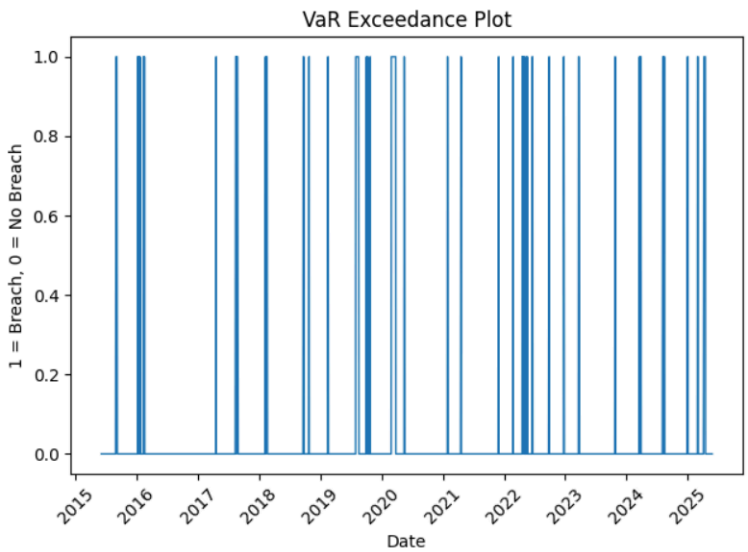



## 3) Horizon 10 No KDE on Tails

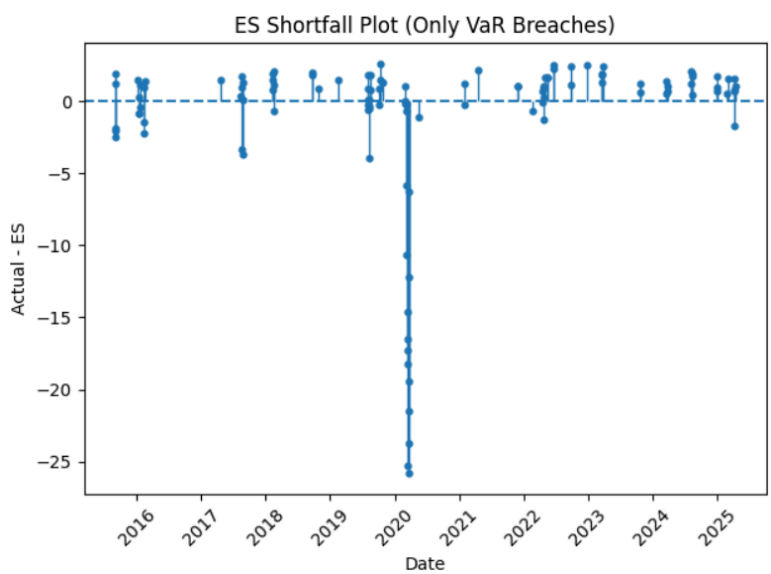
### a)Step Size 1 and 2 assets

| Count of Reason | Column Labels | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | lower | | | lower Total | upper | | | upper Total | Grand Total |
| Row Labels | invalid | sparse | (blank) | | invalid | sparse | (blank) | | |
| INFY.NS | | | | | 12 | 93 | | 105 | 105 |
| SBIN.NS | 435 | 152 | | 587 | | 37 | | 37 | 624 |
| **Grand Total** | **435** | **152** | | **587** | **12** | **130** | | **142** | **729** |

**Exceedence Plot**




**ES Shortfall plot**

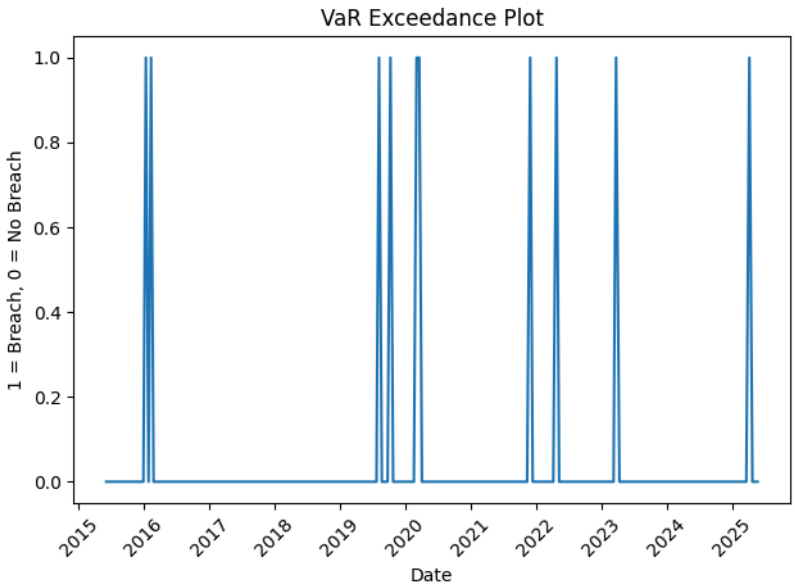



**Here we can see that points are clustering because there will be 9 overlapping days with rolling window .**

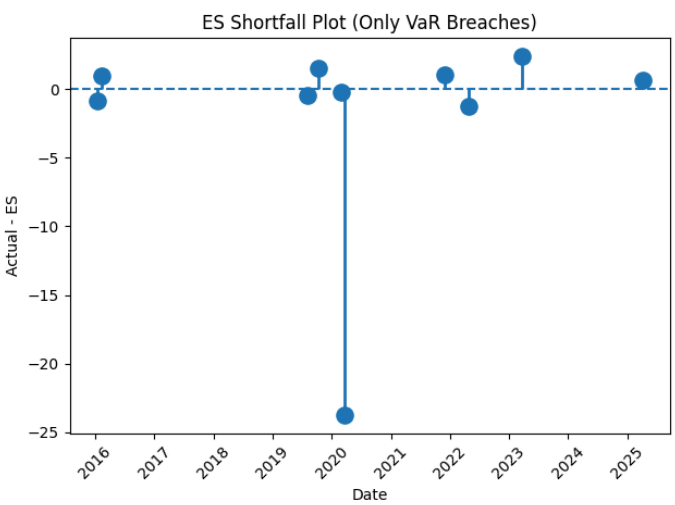
**b)Step size 10 and 2 assets**

| Count of Reason | Column Labels | | | | | | |
|---|---|---|---|---|---|---|---|
| | lower | | | lower Total | upper | | upper Total | Grand Total |
| Row Labels | invalid | sparse | (blank) | | invalid | sparse | (blank) | | |
| INFY.NS | | | | | 1 | 10 | | 11 | 11 |
| SBIN.NS | 43 | 15 | | 58 | | 4 | | 4 | 62 |
| **Grand Total** | **43** | **15** | | **58** | **1** | **14** | | **15** | **73** |

## Exceedence Plot



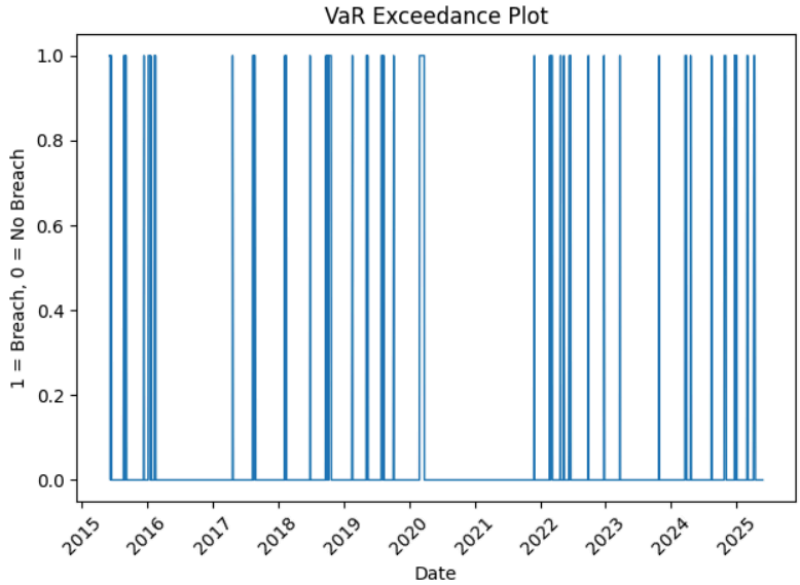

## ES Shortfall plot



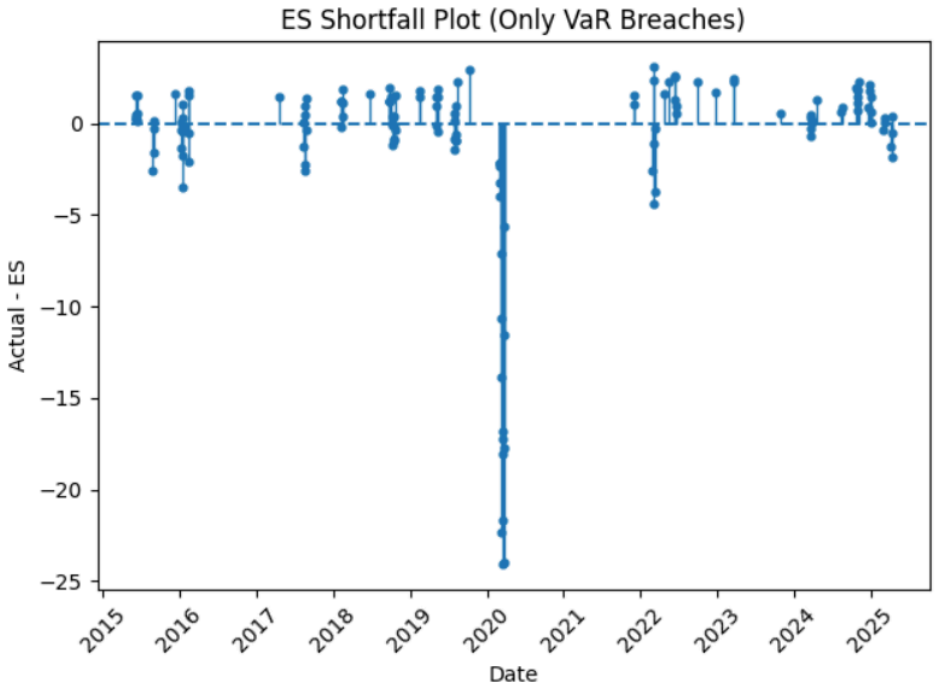

## c) Step size 1 and 4 assets

| Count of Reason | Column Labels | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | lower | | | lower Total | upper | | | upper Total | Grand Total |
| Row Labels | invalid | sparse | (blank) | | invalid | sparse | (blank) | | |
| HINDUNILVR.NS | 535 | 1288 | | 1823 | | 344 | | 344 | 2167 |
| INFY.NS | | | | | 13 | 95 | | 108 | 108 |
| SBIN.NS | 435 | 152 | | 587 | | 37 | | 37 | 624 |
| TATAMOTORS.NS | 68 | 111 | | 179 | 24 | 741 | | 765 | 944 |
| **Grand Total** | **1038** | **1551** | | **2589** | **37** | **1217** | | **1254** | **3843** |

**Exceedence Plot**



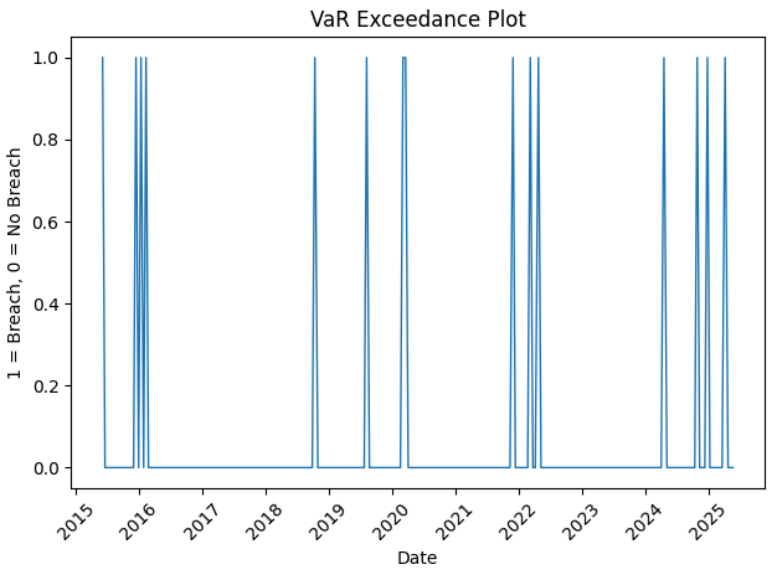

**ES Shortfall plot**



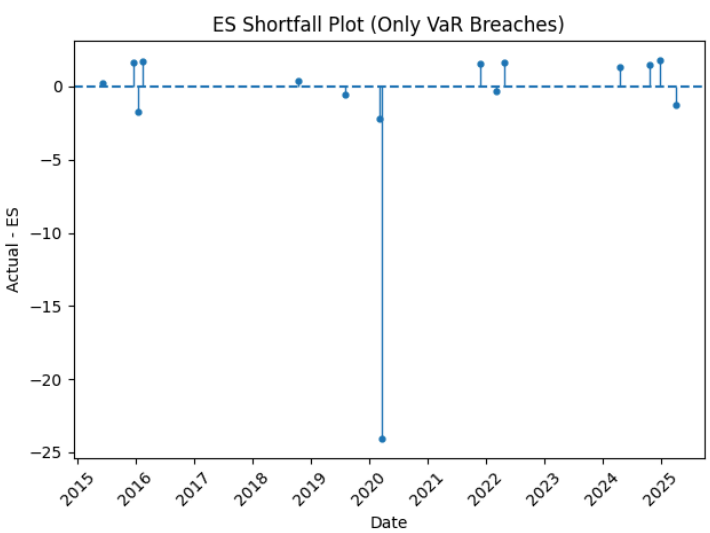

**d) Step size 10 and 4 assets**

| Count of Reason | Column Labels | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | lower | | | lower Total | upper | | | upper Total | Grand Total |
| Row Labels | invalid | sparse | (blank) | | invalid | sparse | (blank) | | |
| HINDUNILVR.NS | 52 | 131 | | 183 | | 34 | | 34 | 217 |
| INFY.NS | | | | | 1 | 10 | | 11 | 11 |
| SBIN.NS | 43 | 15 | | 58 | | 4 | | 4 | 62 |
| TATAMOTORS.NS | 10 | 13 | | 23 | 4 | 74 | | 78 | 101 |
| **Grand Total** | **105** | **159** | | **264** | **5** | **122** | | **127** | **391** |

## Exceedence Plot




## ES Shortfall plot

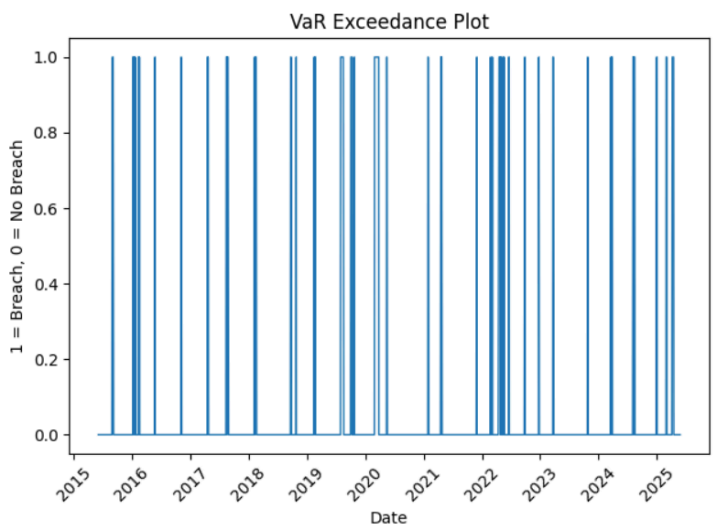


## 4) Horizon 10 with KDE on Tails

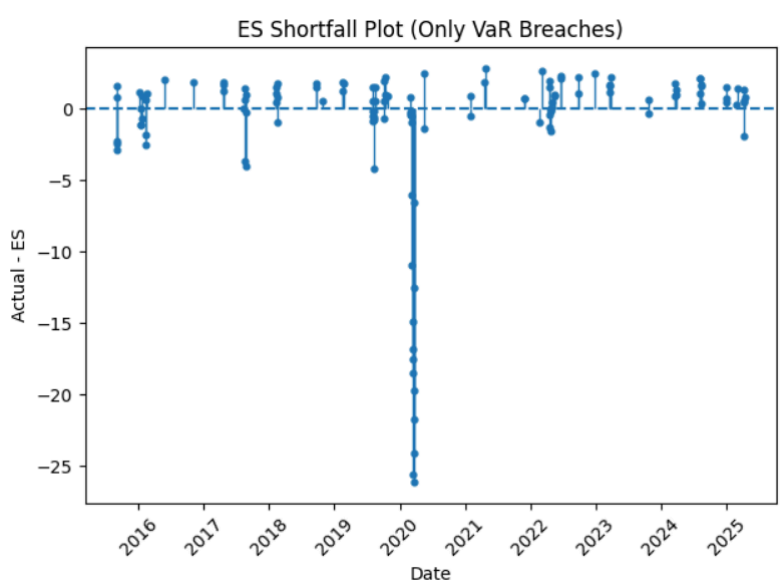
### a)Step Size 1 and 2 assets

| Count of Reason | Column Labels | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | lower | | | lower Total | upper | | | upper Total | Grand Total |
| Row Labels | invalid | sparse | (blank) | | invalid | sparse | (blank) | | |
| INFY.NS | | | | | 12 | 93 | | 105 | 105 |
| SBIN.NS | 435 | 152 | | 587 | | 37 | | 37 | 624 |
| **Grand Total** | **435** | **152** | | **587** | **12** | **130** | | **142** | **729** |

### Exceedence Plot



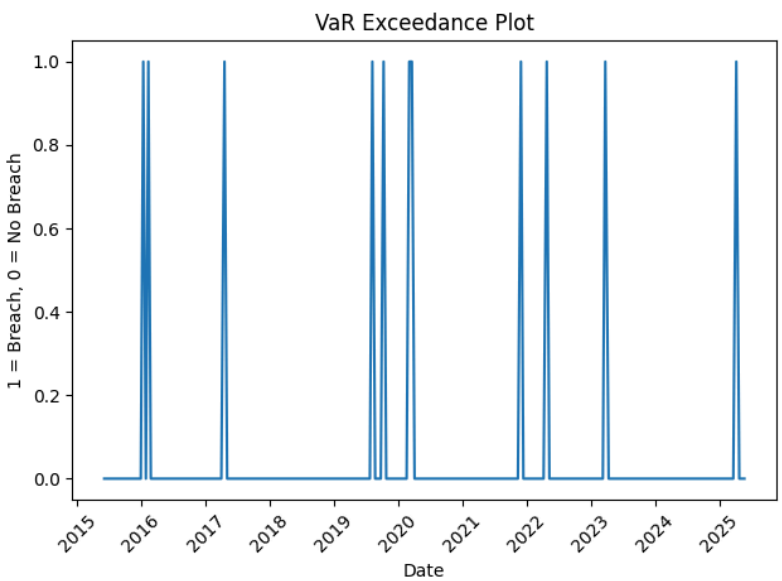

### ES Shortfall plot



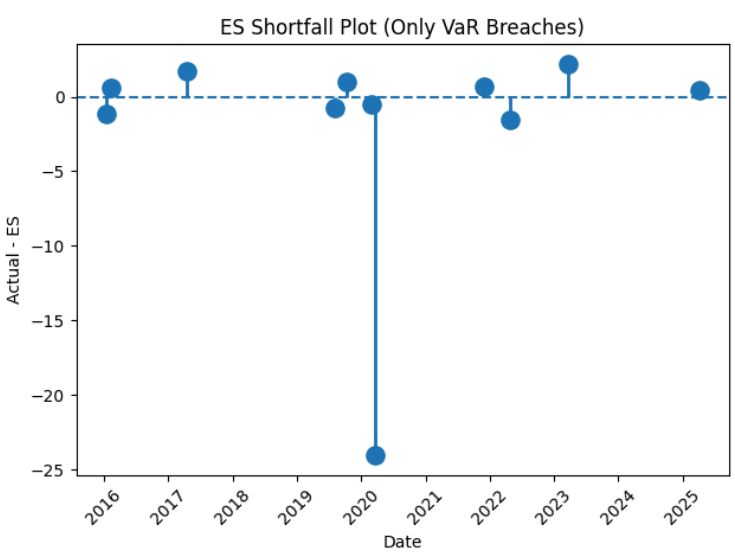
**b)Step size 10 and 2 assets**

| Count of Reason | Column Labels | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | lower | | | lower Total | upper | | | upper Total | Grand Total |
| Row Labels | invalid | sparse | (blank) | | invalid | sparse | (blank) | | |
| INFY.NS | | | | | 1 | 10 | | 11 | 11 |
| SBIN.NS | 43 | 15 | | 58 | | 4 | | 4 | 62 |
| **Grand Total** | **43** | **15** | | **58** | **1** | **14** | | **15** | **73** |

**Exceedence Plot**



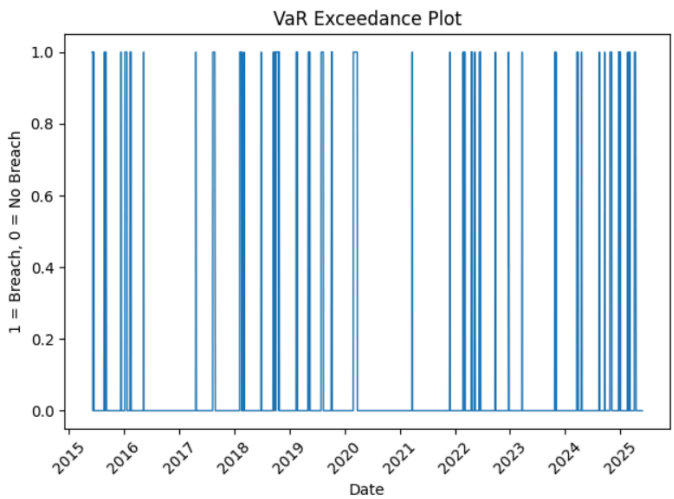

**ES Shortfall plot**



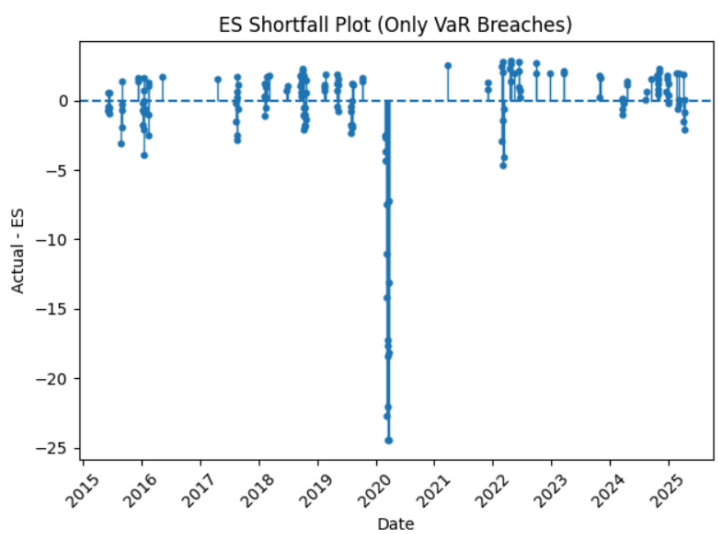
### c) Step size 1 and 4 assets

| Count of Reason | Column Labels | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | lower | | | lower Total | upper | | | upper Total | Grand Total |
| Row Labels | invalid | sparse | (blank) | | invalid | sparse | (blank) | | |
| HINDUNILVR.NS | 535 | 1288 | | 1823 | | 344 | | 344 | 2167 |
| INFY.NS | | | | | 13 | 95 | | 108 | 108 |
| SBIN.NS | 435 | 152 | | 587 | | 37 | | 37 | 624 |
| TATAMOTORS.NS | 68 | 111 | | 179 | 24 | 741 | | 765 | 944 |
| **Grand Total** | **1038** | **1551** | | **2589** | **37** | **1217** | | **1254** | **3843** |

### Exceedence Plot



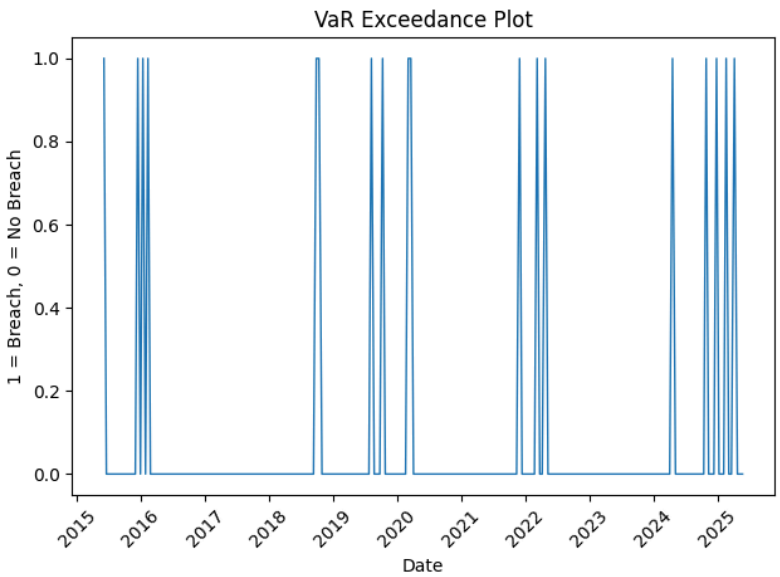

### ES Shortfall plot



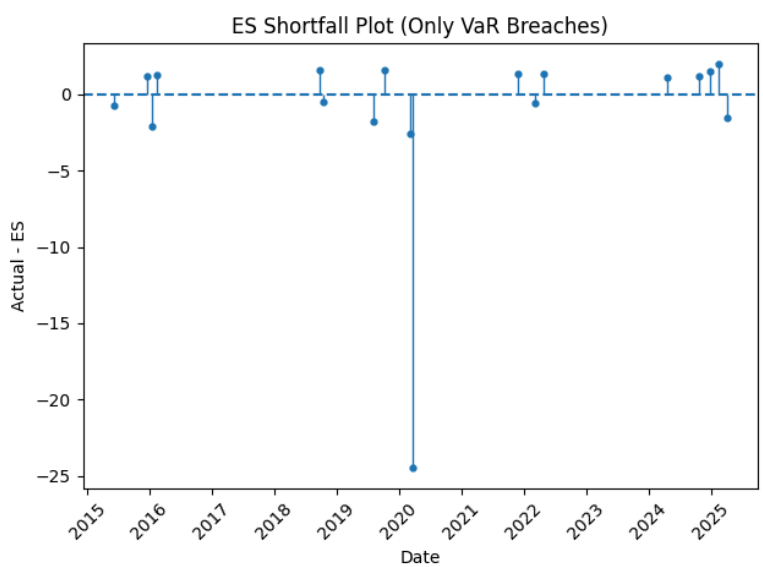

### d) Step size 10 and 4 assets

| Count of Reason | Column Labels | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | lower | | | lower Total | upper | | | upper Total | Grand Total |
| Row Labels | invalid | sparse | (blank) | | invalid | sparse | (blank) | | |
| HINDUNILVR.NS | 52 | 131 | | 183 | | 34 | | 34 | 217 |
| INFY.NS | | | | | 1 | 10 | | 11 | 11 |
| SBIN.NS | 43 | 15 | | 58 | | 4 | | 4 | 62 |
| TATAMOTORS.NS | 10 | 13 | | 23 | 4 | 74 | | 78 | 101 |
| **Grand Total** | **105** | **159** | | **264** | **5** | **122** | | **127** | **391** |

## Exceedence Plot




## ES Shortfall plot



For step size 1 and horizon 10 case we cannot rely on any backtests so no need of interpreting christophersson p value being zero .

From the results we can observe that :

1) If we consider backtests then dq p for 4 lags is less than 0.05( for horizon 10 step size 1 and 4 assets case) for with kde on tails method , which indicates considering without kde on tails method is better

2) But except in one case(horizon is 1 and step size 10 , assets =2) almost all cases, with kde on tails has lesser FZ0 score than without kde on tails and also in one case(horizon is 1 and step size 1 , assets =4) without kde on tails has mean fZ0 score greater than baseline (However median FZ0 score is lesser than baseline so it may be fine ) which suggests to prefer with kde on tails method than without kde on tails . and also the gap between baseline and with kde on tails is around 0.2 whereas for baseline and without kde is 0.1 for horizon 10 and the gap is almost same for horizon 1

So FZ0 score wise with kde on tails is better and backtests wise without kde on tails is better and backtests is what it matters more and also as median FZ0 score is lesser than baseline for with out kde on tails in all case so considering without kde on tails is better